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INSTITUTE OF TECHNOLOGY OF CAMBODIA

Internet Programming 1

Project Proposal

Sakol: A University Exploration and Major Guidance Web Application for Cambodian Students

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1. Introduction

1.1 Background and Problem Statement

Choosing a university and a major is an important decision for students because it can influence their future education and career opportunities. However, students may find it difficult to make these decisions because information about universities, majors, and scholarships is often spread across different sources.

Students may also be unsure about which major matches their interests, strengths, and career goals. Without an organized source of information and guidance, they may spend a lot of time searching or struggle to understand the differences between available majors.

To address these problems, this project proposes **Sakol**, a web application that centralizes higher education information and provides major guidance for Cambodian students. The platform will allow students to explore universities, majors, and scholarships in one place while using assessments to better understand their interests and preferences.

Sakol will support students in exploring their options and making informed decisions. It will not choose a major for them or guarantee that a recommended major is the best choice.

1.2 Project Objectives

The main objective of **Sakol** is to centralize higher education information and provide guidance that helps Cambodian students explore majors and make more informed decisions about their future studies.

The specific objectives are:

- To collect and organize information about universities, majors, and scholarships in one accessible platform.
- To help students search for and compare different study options.
- To provide a Find My Major assessment based on students' interests, strengths, and career preferences.
- To provide an MBTI-inspired personality assessment that helps students reflect on their personality preferences.

- To offer major suggestions with explanations so students can explore options that may interest them.
- To allow registered users to save their favorite universities and majors and review their previous assessment results.
- To provide an administration system for maintaining information and assessment questions.

1.3 Proposed Solution

Sakol will be developed as a web application that students can access through a web browser. It will bring university, major, and scholarship information together in one place and provide two assessments to support students in exploring suitable study options.

The first assessment, Find My Major, will focus on students' interests, strengths, and career preferences. The second assessment will be an MBTI-inspired questionnaire that helps students reflect on four personality preference dimensions.

The application will use a scoring system to generate major recommendations and explain why certain majors may be suitable for a student's answers. These recommendations will be provided as guidance rather than a final decision about the student's future.

The project will use Vue 3 and TypeScript for the frontend, NestJS for the backend, PostgreSQL for the database, and Prisma as the database ORM.

2. Overall Description and Project Scope

2.1 Target Users

The main target users of Sakol are Cambodian high school students, university applicants, and students who want to explore different majors.

The main user groups are:

- **Students and visitors:** Explore universities, majors, and scholarships.
- **Registered users:** Can access all visitor features, complete the Find My Major and MBTI-inspired assessments, view their results, save favorite universities and majors, and review their assessment history.

- **Administrators:** Add, update, and delete university, major, and scholarship information, as well as manage assessment questions and related settings.

The application will initially focus on students who need help exploring higher education options in Cambodia.

2.2 Main Features and Project Boundaries

The project will include the following main features.

- **University Directory:** Users can browse universities and view details such as the university name, location, description, website, and available majors when the information is available.
- **Major Directory:** Users can explore different majors and read descriptions about their study areas, related subjects, and possible career paths.
- **Scholarship Directory:** Users can browse scholarship information, including eligibility requirements, benefits, deadlines, and application links when available. Scholarship details will depend on the information collected and verified for the project.
- **Find My Major Assessment:** Users answer questions about their interests, strengths, preferred activities, and career preferences. The system calculates scores and suggests majors based on the answers.
- **MBTI-Inspired Personality Assessment:** Users complete a questionnaire based on four personality preference dimensions: Extraversion–Introversion, Sensing–Intuition, thinking–Feeling, and Judging–Perceiving. The system provides a four-letter preference profile and a simple explanation. This will be an original, simplified educational self-reflection assessment inspired by these dimensions. It will not claim to be an official MBTI assessment or a psychological diagnosis.
- **Combined Major Recommendations:** The system will use the results of both assessments to provide additional major suggestions and explanations. Students will be encouraged to consider their interests, abilities, academic requirements, and personal circumstances before making a decision.
- **User Accounts and Saved Items:** Registered users can create an account, log in, save favorite universities and majors, and review their previous assessment results.
- **Administration System:** Administrators can manage university, major, and scholarship information and maintain the questions and scoring mappings used by the assessments.

2.3 Assumptions and Limitations

The project assumes that students can access the application through a modern web browser and an internet connection. It also assumes that the collected sufficient university, major, and scholarship information from reliable sources.

Some informations may be incomplete or change over time. Therefore, the accuracy of the directory will depend on the availability and verification of the collected data.

The assessment results will depend on the questions, scoring rules, and major mappings designed for the application. They will not guarantee academic success or a suitable career. The system is intended to support exploration, not replace advice from teachers, career counselors, universities, or students' own judgment.

As this is a semester project, the number of universities, majors, scholarships, and assessment questions may be limited by the available development time.

3. System Requirements

3.1 User Requirements and User Stories

Sakol is designed to help students explore higher education opportunities and receive guidance when choosing a major.

- As a visitor, I want to explore universities, majors, and scholarships so that I can find relevant information in one place.
- As a registered user, I want to complete assessments and receive major suggestions so that I can better understand my interests and preferences.
- As a registered user, I want to view recommendation explanations, save favorite items, and review previous assessment results so that I can make more informed decisions.
- As an administrator, I want to manage directory information, assessment questions, and scoring rules so that the platform remains accurate and useful.

3.2 Functional Requirements

The system shall provide the following functions.

FR-01: University Management and Search

- The system shall display a list of universities.
- Users shall be able to search for universities by name and filter by available criteria, such as location.
- Users shall be able to view the details of a selected university.
- Administrators shall be able to create, view, update, and delete university records.

FR-02: Major Management and Search

- The system shall display a list of majors with their descriptions.
- Users shall be able to search for majors by name or related study area.
- Users shall be able to view details about a major, including related career paths when available.
- Administrators shall be able to manage major records and their relationships with universities where applicable.

FR-03: Scholarship Directory

- The system shall display available scholarship records.
- Users shall be able to view scholarship details and eligibility requirements when provided.
- Administrators shall be able to add, update, and remove scholarship information.

FR-04: User Registration and Login

- Users shall be able to register an account and log in.
- The system shall validate required registration and login information.
- The system shall restrict account-specific functions to authenticated users.
- The system shall provide a secure way to manage user sessions.

FR-05: Find My Major Assessment

- The system shall display questions about interests, strengths, activities, and career preferences.

- Users shall be able to submit their answers.
- The system shall calculate scores using predefined scoring rules.
- The system shall generate a ranked list of suggested majors based on those scores.
- The system shall provide explanations for the recommendations.

FR-06: MBTI-Inspired Personality Assessment

- The system shall display questions related to four personality preference dimensions.
- The system shall calculate the user's preference on each dimension based on their answers.
- The system shall generate a four-letter preference profile and a simple explanation.
- The system shall clearly state that the assessment is an educational self-reflection tool, not an official MBTI assessment or psychological diagnosis.

FR-07: Combined Major Recommendations

- The system shall use results from the two assessments to provide additional major suggestions.
- The system shall explain how the assessment results relate to the suggested majors.
- The system shall allow users to explore the recommended majors in the major directory.
- The system shall present results as guidance rather than guaranteed predictions.

FR-08: Saved Items and Assessment History

- Registered users shall be able to save and remove favorite universities and majors.
- Registered users shall be able to view their saved items.
- The system shall store assessment attempts and results for authenticated users.
- Registered users shall be able to review their previous assessment results.

FR-09: Administrative Functions

- The system shall provide administrator-only access to content management functions.
- Administrators shall be able to manage universities, majors, scholarships, assessment questions, and scoring mappings.
- The system shall prevent ordinary users from accessing administrative functions.

FR-10: Validation and Data Persistence

- The system shall validate user inputs before processing or storing them.
- The system shall display understandable error messages when an operation fails.
- The system shall store application data in the PostgreSQL database.
- The system shall preserve relationships between users, saved items, assessment attempts, questions, responses, and results.

3.3 Non-Functional Requirements

- **Usability:** The interface should be clear and easy to understand for students with different levels of technical experience.
- **Performance:** Common actions, such as opening a directory, searching for a major, and viewing results, should respond within a reasonable time under normal project testing conditions.
- **Security:** Passwords must not be stored as plain text. Authentication and authorization shall protect user accounts and administrator functions. The backend shall validate incoming data and restrict access to protected operations.
- **Data Integrity:** The system shall use appropriate database constraints and relationships to reduce invalid, duplicate, or inconsistent records.
- **Reliability:** The application should handle invalid input and common errors without crashing and should provide clear feedback to users.
- **Maintainability:** The source code should be organized into understandable modules and components to make future changes easier.
- **Compatibility and Responsiveness:** The application should work with modern desktop and mobile browsers and adapt to different screen sizes.
- **Testability:** Important functions, including authentication, searching, assessment scoring, saving items, and access permissions, should be tested.
- **Privacy and Transparency:** Users should understand the purpose and limitations of the assessments. Personal information and assessment history should be accessible only through appropriate permissions.

3.4 User Roles and Access Permissions

The application will use three main access levels.

Function	Visitor	Registered User	Administrator
Browse universities, majors, and scholarships	Yes	Yes	Yes
Search and view directory details	Yes	Yes	Yes
Complete Find My Major assessment	No	Yes	Yes, for testing purposes
Complete MBTI-inspired assessment	No	Yes	Yes, for testing purposes
View personal assessment results and history	No	Yes	Only their own results as a user
Save and manage favorite items	No	Yes	Yes, as a user
Manage directory information	No	No	Yes
Manage assessment questions and scoring rules	No	No	Yes

Visitors who want to take an assessment will be asked to register or log in first. After successful authentication, they can complete the assessment and view their results.

4. System Design and Technology

4.1 Technology Stack and System Architecture

The project will use the following technologies.

Technology	Purpose
Vue 3	Building the user interface
TypeScript	Adding types to make the code easier to understand and maintain
NestJS	Developing backend APIs and application logic
PostgreSQL	Storing application data in a relational database
Prisma ORM	Defining database models and accessing the database
Git and GitHub	Tracking code changes and maintaining the project repository

The application will follow a client-server architecture. The Vue frontend will provide the pages and forms that users interact with. It will send requests to the NestJS backend through APIs.

The backend will handle authentication, input validation, directory operations, assessment scoring, recommendation generation, and access permissions. Prisma will connect the backend to PostgreSQL, where application records will be stored. This separation will help keep the interface, business logic, and database operations organized.

4.2 Database Design

The database will be designed to store directory information, user accounts, saved items, and assessment data.

The initial database entities are:

- **User:** Stores account information and user roles.
- **University:** Stores university names, descriptions, locations, and other available details.
- **Major:** Stores major names, descriptions, study areas, and related career information.
- **Scholarship:** Stores scholarship descriptions, eligibility requirements, benefits, deadlines, and application links when available.
- **SavedItem:** Records the universities or majors saved by registered users.
- **Assessment:** Defines an assessment type, such as Find My Major or the MBTI-inspired assessment.
- **Question:** Stores assessment questions and their associated dimensions or scoring categories.
- **AnswerOption:** Stores possible answers and their associated scoring values.
- **AssessmentAttempt:** Records each assessment attempt by a user, including its completion status and date.
- **AssessmentResponse:** Stores the answers submitted for each assessment attempt.
- **AssessmentResult:** Stores the calculated results and recommendations from an assessment attempt.
- **MajorAssessmentMapping:** Defines how assessment categories or scores relate to majors.

The entities will be connected using primary keys, foreign keys, and appropriate relationships. For example, one user can have multiple saved items and assessment attempts. Each assessment attempt can contain multiple responses and produce a result.

The final database design will be refined using an entity-relationship diagram (ERD) before implementation. The design will aim to reduce unnecessary duplication and maintain data consistency.

4.3 Interface Design

The interface will focus on simplicity and ease of use. Students should be able to find information without needing to understand complicated menus or technical terms.

The planned pages are:

- **Home:** Introduces Sakol and provides links to its main features.
- **Universities:** Displays searchable university information.
- **University Details:** Shows information about a selected university.
- **Majors:** Displays available majors and search options.
- **Major Details:** Explains a selected major and its possible career paths.
- **Scholarships:** Displays scholarship information.
- **Find My Major:** Presents assessment questions and recommendations.
- **Personality Assessment:** Presents the MBTI-inspired questionnaire and results.
- **Combined Guidance:** Shows major suggestions using both assessment results.
- **Login and Registration:** Allows users to access an account.
- **Saved Items and Assessment History:** Displays saved content and previous results for registered users.
- **Admin Dashboard:** Provides content and assessment management functions.

The pages will use reusable Vue components where possible. The layout will be responsive so that users can access the application on desktop computers, tablets, and mobile devices.

4.4 Assessment and Recommendation Design

4.4.1 Find My Major Assessment

The Find My Major assessment will focus on three areas:

1. **Interests:** What activities and subjects the student enjoys.
2. **Strengths:** What skills or types of tasks the student feels confident performing.

3. **Career Preferences:** What kinds of work environments, responsibilities, or future activities the student prefers.

Each answer will contribute points to one or more predefined categories associated with majors. After the student submits the assessment, the system will calculate the scores and rank the majors based on their matching scores.

The results page will show a list of suggested majors and explain the factors that contributed to each suggestion. Students will also be able to open the major details page to learn more.

The scoring rules will be designed to provide consistent results and will be adjusted through testing. The assessment will not claim to measure a student's abilities objectively or predict future success.

4.4.2 MBTI-Inspired Personality Assessment

The personality assessment will be based on four preference dimensions:

- **Extraversion (E) and Introversion (I):** Preferences related to social interaction and how a person directs their attention.
- **Sensing (S) and Intuition (N):** Preferences related to concrete information and broader patterns or possibilities.
- **Thinking (T) and Feeling (F):** Preferences related to logical analysis and consideration of values or people when making decisions.
- **Judging (J) and Perceiving (P):** Preferences related to structure and planning versus flexibility and openness.

The system will calculate a preference for each dimension based on the user's answers and produce a four-letter profile. It will then display a short explanation of the profile and encourage the student to reflect on their preferences.

This assessment will be a simplified educational questionnaire inspired by the four dimensions. It will not be presented as an official MBTI instrument, and it will not diagnose personality or determine which major a student must choose.

4.4.3 Combined Major Recommendations

After completing both assessments, users will be able to view combined guidance. The system will compare the results from Find My Major with the personality preference profile and use predefined mappings to suggest majors for further exploration.

The combined result may consider:

- The student's interest and strength scores.
- The student's career preferences.
- The student's personality preference profile.
- The relationship between these results and the characteristics of different majors.

The system will provide explanations for its suggestions rather than showing only a list of major names. If the two assessments produce different suggestions, the application can present both sets of results and explain that students may have several suitable options.

The combined guidance will be treated as a supporting tool. Students should also consider university admission requirements, tuition fees, scholarship opportunities, academic performance, and their own goals.

5. Implementation and Testing Plan

5.1 Development Approach

The project will be developed step by step throughout the semester. Each stage will focus on a small group of features so that progress can be checked and problems can be corrected early.

The planned development stages are:

1. **Requirements and Planning:** Confirm the project scope, identify user requirements, prepare the system requirements, and design the initial ERD.
2. **Project Setup and Database:** Set up Vue, TypeScript, NestJS, PostgreSQL, and Prisma. Create the initial database models and connect the backend to the database.

3. **Directory Features:** Implement university, major, and scholarship pages, including searching and viewing details.
4. **Authentication and User Features:** Implement registration, login, saved items, and assessment history.
5. **Assessment Development:** Build the Find My Major assessment, scoring rules, personality assessment, and result pages.
6. **Administration:** Implement protected functions for managing directory content, questions, and scoring mappings.
7. **Testing and Improvement:** Test the main functions, fix errors, improve usability, and prepare the final project documentation.

Git and GitHub will be used to track changes and keep a history of development. The project will be reviewed regularly to make sure the work remains consistent with the requirements.

5.2 Semester Timeline

The project is planned for development from Week 2 to Week 16.

Period	Planned Activities
Weeks 2–3	Define requirements, write the proposal, and prepare the initial ERD
Weeks 4–5	Set up the project, database, and basic backend APIs
Weeks 6–7	Build university and major directories and search functions
Weeks 8–9	Add scholarship information, authentication, and saved items
Weeks 10–11	Develop Find My Major questions, scoring, and recommendations
Weeks 12–13	Develop the personality assessment and combined guidance
Weeks 14–15	Complete administration functions, testing, and improvements
Week 16	Final testing, documentation, and project completion

- This timeline is a starting plan. Some tasks may overlap, and adjustments may be necessary depending on development progress and feedback during the semester.

5.3 Testing Strategy

- **Functional Testing:** Check whether features work as required, including searching, viewing details, registration, login, saving items, and viewing assessment results.
- **Assessment Testing:** Use different sets of answers to verify that scoring rules produce consistent results, that rankings are calculated correctly, and that explanations match the configured mappings.
- **API and Database Testing:** Check that backend APIs return the expected results, database relationships work correctly, and records can be created, updated, retrieved, and deleted as permitted.
- **Validation and Error Testing:** Submit missing, invalid, or incorrect information to confirm that the system responds with clear error messages and does not save invalid data.
- **Security Testing:** Verify that protected functions require the correct permissions, passwords are handled securely, and ordinary users cannot access administrative operations.
- **Usability and Compatibility Testing:** Ask a small number of users to try the main pages and assessments, gather feedback, and check the interface on different screen sizes and modern browsers.

Any problems found during testing will be recorded, prioritized, and corrected before the final submission.

6. Risks and Mitigation

Several risks may affect the development of **Sakol**.

Risk	**Possible Impact**	**Mitigation**
Incomplete or outdated university and scholarship information	Users may receive incorrect information	Collect information from reliable sources and record source links and relevant dates where possible

Limited development experience	Some features may take longer than expected	Build the application in small stages and prioritize essential features
Assessment scoring is not sufficiently accurate	Recommendations may not match students' expectations	Define clear scoring rules, test different answer patterns, and improve mappings based on feedback
The project scope becomes too large	Important features may remain unfinished	Focus on core features first and postpone non-essential functions
Authentication or access-control problems	User information or administrative functions may be exposed	Follow secure development practices and test permissions
Database design problems	Data may become inconsistent or difficult to maintain	Prepare and review the ERD before implementing the full database
Limited time for testing	Errors may remain in the final application	Test each feature as it is developed and reserve the final weeks for improvements

The project will prioritize a working and testable core application over adding too many features. This approach will help ensure that the most important requirements can be completed within the semester.

7. Expected Outcomes

At the end of the semester, the expected outcome is a working web application that helps Cambodian students explore universities, majors, and scholarships and receive basic guidance when considering their future studies.

The completed application is expected to provide:

- A searchable directory of universities and majors.

- A scholarship directory containing available and verified information.
- A Find My Major assessment with scoring and ranked recommendations.
- An MBTI-inspired personality assessment with a four-letter preference profile.
- Combined major guidance based on the results of both assessments.
- User registration, login, saved items, and assessment history.
- An administration system for maintaining directory information and assessment settings.
- A structured database and backend API that support the main application features.
- Documentation and tests demonstrating how the system meets its requirements.

In addition to the application itself, this project will provide practical experience in frontend development, backend API development, relational database design, authentication, software testing, and project organization.

8. Conclusion

Sakol aims to make the process of exploring higher education options more convenient for Cambodian students by bringing university, major, and scholarship information together in one platform. Its assessment features will provide additional guidance by helping students reflect on their interests, strengths, career preferences, and personality preferences.

The project will be developed using Vue 3, TypeScript, NestJS, PostgreSQL, and Prisma. By following an SRS-based planning process and implementing the application in stages, the project will focus on clear requirements, manageable development tasks, and regular testing.

Although the assessment results cannot determine the perfect major for every student, Sakol is intended to help students discover options, understand their preferences, and make more informed decisions about their education.

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